

Project Design Phase
Proposed Solution Template

Date	3 October 2026
Team ID	
Project Name	
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Users frequently struggle to stay within strict financial budgets when planning room decor, organizing parties, or buying jewelry due to scattered pricing across multiple e-commerce platforms and a lack of real-time itemized cost calculation tools.
2.	Idea / Solution description	PocketSmart AI is an interactive Streamlit web dashboard integrated with Google Gemini 3.8 Flash (gemini-3.8-flash via the google-gemini SDK). It provides automated, cross-platform price recommendations across four core scenarios: Home Interior Planning (IKEA, Amazon), Event & Party Packages (Swiggy, Zomato, OYO), Jewelry Styling (Amazon, Flipkart), and an integrated Core Expense Ledger with real-time balance metrics and interactive Plotly visual analytics.
3.	Novelty / Uniqueness	Unlike standard shopping platforms or static budgeting apps, PocketSmart AI combines generative AI scenario planning directly with an active financial tracking ledger. It enforces strict budget caps before purchases occur, generates itemized cross-platform bundles dynamically, and provides an all-in-one experience without requiring users to manually toggle between multiple shopping tabs.
4.	Social Impact / Customer Satisfaction	Social Impact: Promotes financial literacy, responsible spending, and debt reduction among students, young professionals, and middle-income families.
5.	Business Model (Revenue Model)	Freemium Subscription Model: Offers a free tier for basic AI budget recommendations and expense logging, with a premium subscription tier unlocking unlimited AI prompt runs, multi-budget scenario saving, and custom export options (PDF/Excel financial reports)

6.	Scalability of the Solution	Cloud-Native Architecture: Built using Streamlit and Google Cloud Generative AI (<code>gemini-3.8-flash</code>), allowing serverless deployment that automatically scales to handle high concurrent user traffic with minimal infrastructure overhead. Modular Scenario Expansion: The prompt orchestration engine is designed modularly, making it easy to add new domain categories (such as Travel & Vacation Planning, Wedding Packages, or Educational Expense Planning) without modifying the core codebase. API-Driven Ecosystem: Designed for easy future integration with live e-commerce search APIs and Open Banking APIs to automate real-time price fetching and direct bank transaction syncing.
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